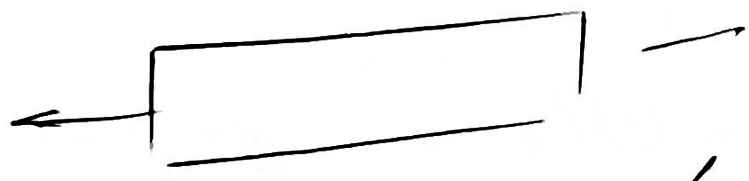
* Stress pattern
  - 3. frictional stress
    - 3.1 @ corner
    - 3.2 @ interior & edge
      - 3.2.1 summer
      - 3.2.2 Winter

frictional stress (only for edge interior, zero for corner)

(summer & winter)

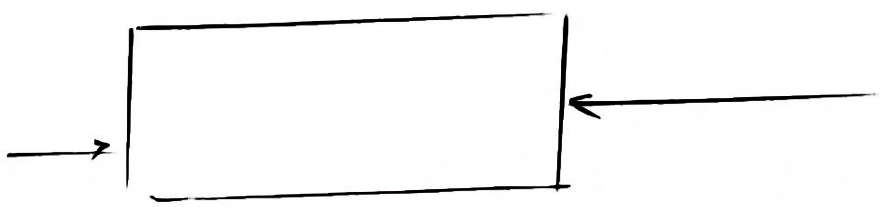
frictional stress is zero @ corner at interior and edge region

I. in summer



friction ke karan fail raha hai toh compressive stres develop hoga @ top & bottom.

II. In winter



Tensile stress develop hoga top & bottom me.

Design of tie bars
- provided where?

Design of tie bar - Provided across Longitudinal joint

16/12

Design of tie bars

As formula

Area of steel for tie bar

① $A_s = \frac{B f s_c}{\sigma_{st}}$

f → 1.5

s_c → 2400 kg/m³

σ_{st} → N/mm² (Mpa) (140 – 175 Mpa)

length

σ_{st} is allowable working tensile stress used in steel in MPa

-Ld formula

② $L_d = \frac{\phi \sigma_{st}}{2 \tau_{bd}}$

ϕ is diameter of bar

τ_d is permissible bond stress of conc.

Class 15 22 Aug 24

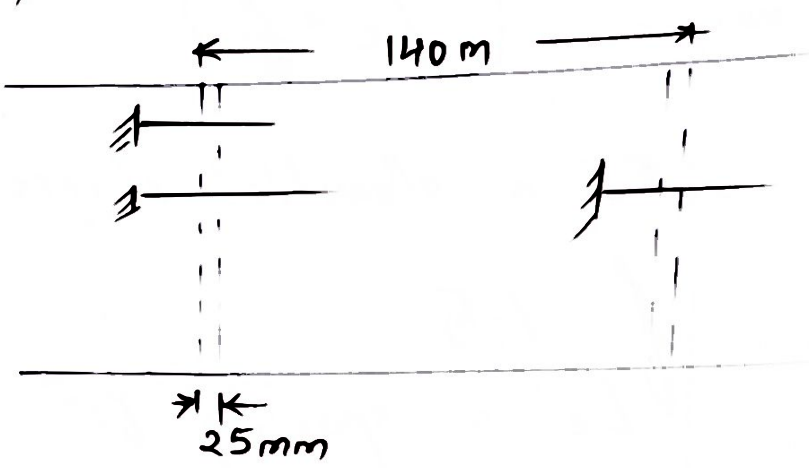
Expansion joint
- purpose
- thickness
- Gap.

Expansion joint

The purpose of expansion joint is to allow the expansion of the pavement due to rise in temperature wrt construction temperature.

Ethickness : 25 mm to 2.5 cm

Gap : 140 m



- kaunsa bar provide kate h?
- kaise dete h dowel bars

Dowel bars are provided in expansion joints.

- Dowel bar is fixed @ one side and free at another side.

- Spacing formula
- $L_c = \frac{\delta}{\alpha [t_2 - t_1]}$

δ is maximum expansion in slab length.

15.2

Contraction jt

- purpose

- Spacing kitne se zyada hai honi chahie

- (i) Pcc slab ki spacing ka formula

- (ii) RCC slab ki spacing ka formula

Contraction joint

The purpose of contraction joint is to allow the contraction of slab due to fall in temperature

distance / spacing ≯ 4.5m

(i) PCC Slab spacing

$$L_c = \left(\frac{2\sigma_c}{\gamma_c * f}\right)$$

σ_c is allowable stress in tension in c/c

$f = 1.5$

L_c = spacing bt^n contraction joints

(ii) for RCC slab.

$$L_c = \left(\frac{2\sigma_c}{\gamma_c * f}\right) * \frac{A_s}{Bh}$$

A_s is area of reinforcement

B is width of slab

h is thickness.

Marshall Stability test.
└ kiske lie kartch?

15.2

Marshall's stability test for mix design for flexible pavement

flowchart of marshalls stab test

Gt

↓

Gm

↓

% air voids V_v

↓

% volume of bitumen V_b

↓

VMA

↓

VFB

15.4

Marshall graphs

- kaunse do increase hote h phir decrease hote h?

- kaunsa decrease hota h

- kaunsa always ↑ ho rha h

- flow value wala kaisa h graph?

- VMA wala kaisa bnta h?

Marshalls graphs

① density vs ⅟age bitumen

Marshalls stability number vs ⅟ bitumen

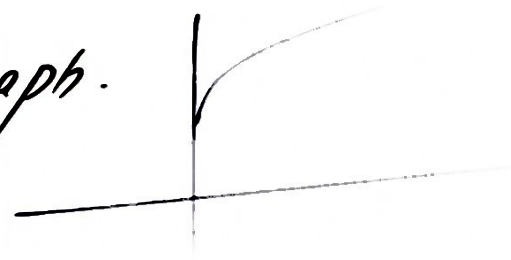
Marshall stab no ↑ till ↓ maximum load value

} increases & then decreases

② ⅟age voids of air v/s ⅟ bitumen - always decreasing graph

③ ⅟age voids filled with bitumin V_{FB} -

always increasing graph.

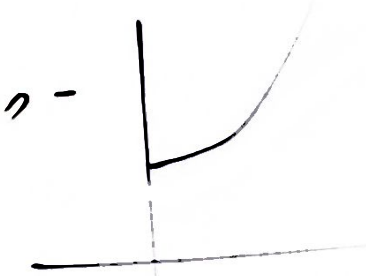


④ flow value vs ⅟ bitumen -

Marshalls max

Flow value means deformation.

or maximum deflection in mm.



⑤ VMA voids in Aggregate vs ⅟ bitumen

inerd

VMA

$V_a + V_b$



⅟ bitumen

Q * Imp one liners.

(Revelling failure occurs in bitumen pavement or flexible pavement is due to - - - - - → open graded aggregate.

Q In hill roads the prevention of landslides for monsoon season is done by - - - - - - → 1) Retaining wall
2) Catch water drain
3) Breast wall

Q 1996 ES In bituminous construction compaction by pneumatic roller also is specified by - - → Pre mix carpet

Q Binders which is recommended in wet and cold climate - - - - - → cutback

JES 1998
Marshall design specification
- MSN (min)
- flow value range
- % air void range
- VFB

Marshall design specification.
(i) Marshall stab no : 340 kg (min)
(ii) flow value [0.25mm unit] : 8 to 17
(iii) % air voids in mix : 3 to 5%.
(iv) Voids filled with bitumen = 75 - 85%.

15. (c) To carry out bituminous patchwork during rainy season most suitable binder is - - - - - - - - - -

→ bituminous emulsion.

Q Bitumen grade 80/100 indicates penetration value = x mm to y mm.

→ 8mm to 10mm

CC road construction steps kya kya h?

- prep
- place
- Mix
- Compct
- Cur

Steps involved in construction of CC roads

1. preparing the subgrade & base course & watering
2. placing the formwork / framework on the prepared base.
3. Mixing & placing of concrete
4. Compaction and floating.
5. Curing

(i) $\frac{80}{100}$ penetration → bituminous road

(ii) $\frac{85}{25}$ blown bitumen → mastic asphalt

(iii) MC - 70 cutback → prime coat.
medium curing

(iv) RT 5 → used in grouting
[Road tar 5]

QES 2009 Which property is not a desirable property of subgrade soil as a highway material. — — → (a) stability
(b) ease of compaction
(c) good drainage
(d) bitumen adhesion

Sequence in increasing order of performance qualities

SD < PMC < SDBM
SD = surface dressing
PMC = premix concrete
SDBM = semidense bituminous macadam
AC = asphaltic concrete
MA = mastic asphalt

Standard bitumen value as per marsh
11 = — — — — 6%

15.9

Railwayeng

- Gauge Defn
- Sizes in mm & inches
  - BG
  - MG
  - NG
  - LG
  - Std gauge
- Selection
  - coc
  - ↓
  - Dvlpmt
  - Phy feat
  - Spd

Railway Engineering

1) Gauge of railway track is defined as clear distance between inner or running faces

(i) Broad gauge : 1.676m 5'6"

(ii) metre gauge : 1m 3'3"

(iii) narrow gauge (NG) 0.762m 2'6"

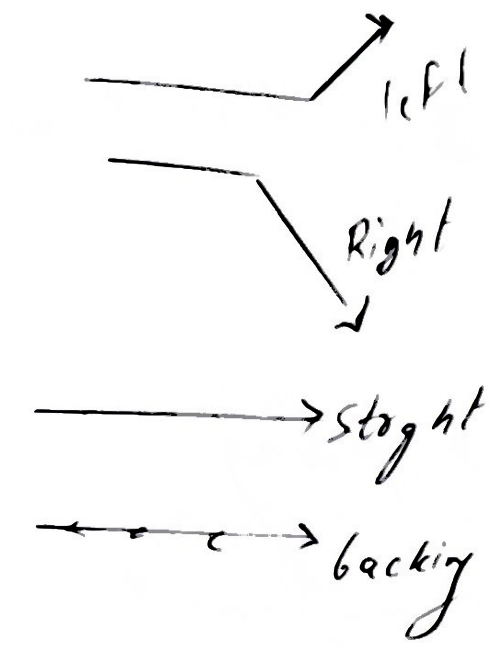
(iv) feather/feeder (LG) gauge : 0.610m 2'

(v) Standard gauge used in delhimetro : 1.435m

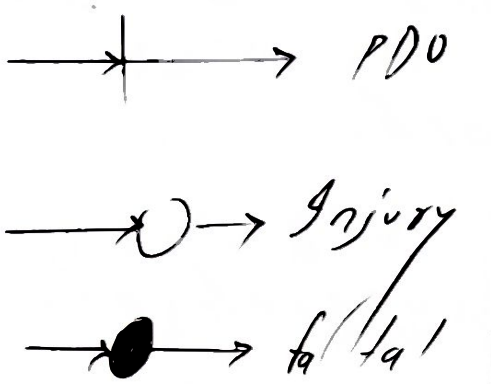
2) Selection of gauge depends on

1. cost of construction.
2. Volume & nature of traffic.
3. Developement of area.
4. Physical features of country
5. Speed of movement.

Vehicle mvmt



Severity



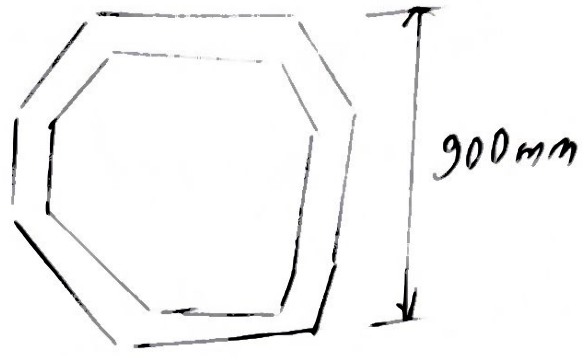
Road surface

C Dry, Clear
W Wet
S Snowy icy
O others

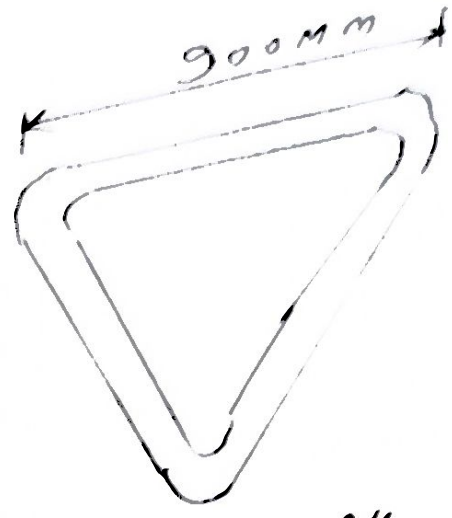
lighting

D Daylight
N Dark no lights
L Dark with Streetlights

Traffic signs



Stop

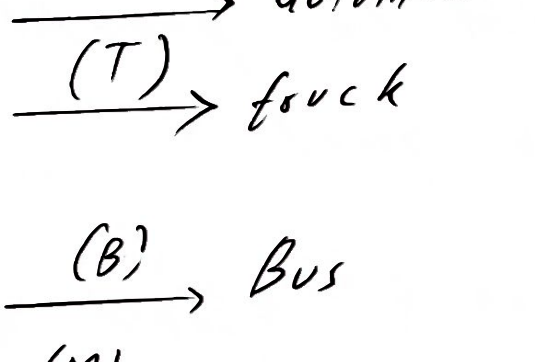


Give way



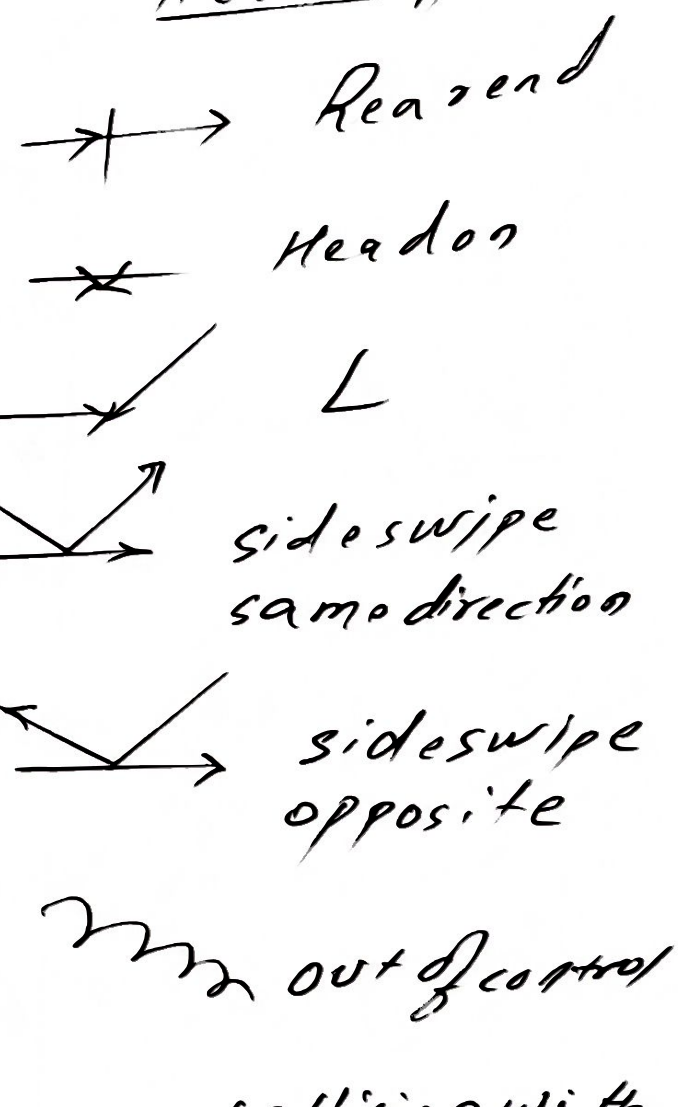
Speed limit

Vehicle type



key to collision diagram symbols

Accident types



Class 16 23 Aug 2023

P-way

for railway or train an ideal P-way is required.

Requirements for an ideal pway

1 Requirement Gauge

1. the gauge should be uniform & ideal.

b level

2. Both the rails should be @ same level in straight track.

c Elevn

3. On curve proper superelevation should be provided to the outer rail.

d str

4. track should have enough lateral strength.

e ecity

5. track must have certain amount of elasticity.

f drng

6. drainage system should be perfect

g renewal joints

7. It should have adequate provision for easy renewal & repair

Coning of wheels

- kyo zaroorath

- level track pe wheel tread dia kaise vary karta h.

- circumference kaisa rehna chahiye str track pe?

& all joints / points & crossing (16.2
should be properly designed.

Coning of wheels

. it is required for the wheel to travel equal distances while travelling in a straight track.

. on the level track the axle may move towards one rail the diameter of wheel tread over the rail increase while it decreases over the other rail.

. on straight & level track the wheel remains in central position & the circumference travelled by both the wheels should be equal.

16.2)

coning of wheels

- advantage wear & tear
- advantage negotiate curve
- advantage central posn.
- Disadvantages
  - man diff
  - edge damage

- it reduces wear & tear of wheel flanges
- it helps vehicle to negotiate curve smoothly.
- it keeps the train in central position

Disadvantage

- manufacturing difficulty
- no base plate are provided hence outer edge of rail gets damaged

Adzing of sleepers

- kya hota h adzing?

Adzing of sleepers

- A groove having an angle of 1 in 20 is being cut on the top of the sleeper so that the rail should be fixed in this location it is called adzing.

Canting of rail
- kyo krte h canting

Canting of rail

in order to minimise the disadvantage due to coning of wheels canting in rail is done. which means the rails are not placed flat but are tilted inward. (slope 1 in 20)

Rail
- kya hota h?

Rail

Rails are member of railway track which are parallel to each other or parallel lines.

- kya krta h?
- it defines a track on which train travels.

- Requirements
  - stl
  - wrng strc
  - foot thick

Requirement of good rails

1 proper composition of steel

2 deep & hard wearing surface of head

3 sufficient foot thickness & foot width

Rail

- type of rail sections
  - BH
  - DH
    - advantage
  - FF
- Rail sections used in india
  - for BG (wt & length wise)
- Metre gauge length & grades
- narrow gauge grade & length

Type of rail sections

① bull headed rail (BH rail)

② Double headed rail (DH rail)

It can be reused after inverting.

③ flat footed rail. (FF rail)

Type of rail sections used in india.

for broad gauge → [IRS] 52 kg/m [13 m]

for metre gauge → 90 R [12 m rail length]
75 R
60 R

for narrow gauge → 50 R [12 m rail length]

Given sleeper density for broad gauge is n+6 if the track laid in 13 m rail length and length of welded rail is 26 m. find out number of sleeper reqd for railtrack length 1690m

Sleeper density calculn

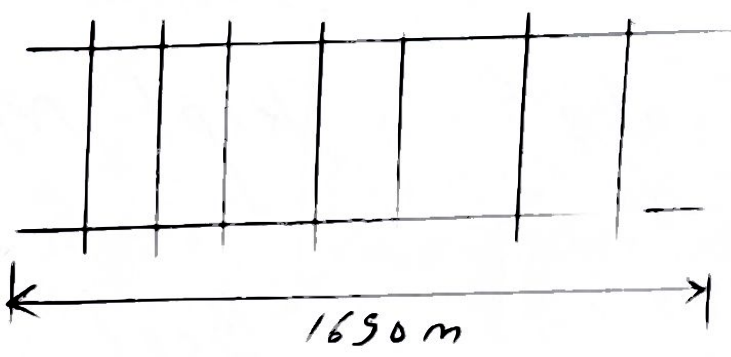
Ans: No sleepers reqd per rail = N+6 = 13+6
= 19 sleeper

* 1 rail length me sleeper reqd = 19 sleepers (13m)

* 1m length me sleeper reqd = 19/13 sleepers

∴ 1690m me sleeper reqd = $1690 \times \frac{19}{13}$

= 2470 sleepers



Types of sleeper

- W
- S
- CI
- CC

Types of sleeper

1. Wooden (Preferred on joints)
   - Hardwood
     - Sal
     - teak (Best)
   - Softwood
     - Chir
     - Deodar
2. Steel
3. Cast iron
4. Concrete.

about 20% tracks of indian railway are steel sleepers.

16.7 Wooden Sleeper

Sizes ?BH

- BG
- MG
- NG

Sleeper density

- kitna hota hai?
- generally?

Size of wooden sleeper

BG → 275 x 25 x 13 cm

MG → 180 x 20 x 11.5 cm

NG → 150 x 18 x 11.5 cm

Sleeper density as per Indian railways is normally kept M+2 to M+7.

Best wood for wooden sleeper → teak

Ballast

- kya hota hai?

Ballast

- Big size aggregate, broken stones are laid & packed below and around sleeper is called ballast

Ballast

- functios:
  - Distribute
  - elastic
  - drng
  - foundn
- Specifions
  - hrd
  - not wear
  - no organic

functions of ballast

1. Distribute the load from sleeper to the base or formation
2. Provides elasticity to the trail
3. Provides good drainage of the track.
4. Provides a suitable foundation for sleepers.

Specifications of good ballast

① it should be hard & durable

② it should not wear out due to physical reasons

③ it should be free from organic & inorganic residues.

16.2

Ballast Type & Size of Sleeper

Ballast sizes Type of sleeper

① 50 mm

② 40 mm

③ 25 m

Wooden sleeper, CI sleeper

CST 9 (central standard trial 9

point & crossing.

Ballast

- Specifics
  - a abrasion val
  - b impact val
  - c Spec grav
  - d Max wtr absptn
  - e elongn & flakiness index

Specification of good ballast

abraision val ≯ 30%.

impact val ≯ 20%.

Specific gravity ≯ 2.65

Max water absptn ≯ 10%.

elongn & flakiness index ≯ 50%.

Wheel se rail me load kaise jata hai

The transfer of wheel load from ~~train~~ wheel to rail is ~~called~~ in form of point load.

Class 17 24 Aug 27 [17.1

Turnout ka section (point & crossing) ka ISI C mein diyah.

Geometric design

- Speed of train
- Martins formula (Safe speed of train)

on transition curve broad gauge & metre gauge: $V = 4.35\sqrt{R-67}$

Velocity : $4.35\sqrt{\text{Radius in m} - 67}$

: Maximum speed based on Railway board formula • speed on Transition curve

(i) for broad gauge = $V = 0.27\sqrt{(C_d + C_a)R}$

(ii) for metre gauge = $V = 0.347\sqrt{(d + C_a)R}$

(iii) for narrow gauge $V = 7.65\sqrt{R-6}$

Max speed upto 50 kmph

100

17.2)

C_a is actual cant / superelevation

C_d is cant deficiency

V is in km/hr

R is in m.

⇌ speed or avg speed

- 1) for $V_m > 50$ kmph

  $V_e = \frac{3}{4} V_m = 0.75 V_m$ } ↓

  $= V_{safe}$

- 2) for $V_m < 50$ kmph

  $V_e = V_m$ } ↓

  $= V_{safe}$

Cant (Superelevation)

- It is the difference btn the height of outer rail to inner rail between on a curve.
- it is same as superelevation term

- Cant
  - inner rail is maintained @ original level & it is also called gradient rail.
- -ve superelevn

Transition curve

- ~~Types~~ Transition curve connects the straight section of train track @ one end and circular curve @ other end.
- Types
  1. Euler's spiral.
     - Ideal transition curve
  2. Cubic spiral
  3. Bernoullis lemniscate
  4. Cubic Parabola
     - Mostly used in indian railway
     - $y = \frac{x^3}{6RL}$
       - x & y are horiz & vert coordinates of length of trans curve

Transn curve

- Types
  - Cubic parabola
    - shift: this is the amount by which a circular curve is shifted inwards ⊥ to meet a transition curve
      - formula: $L^2/24R$
  - Extra widening
    - widening of gauges on curves
    - Reason - Centrifugal force
      - 2) Relative distance travelled by wheel
      - 3) Rigidity of wheel base.
    - $We = W = \frac{13(B+L)^2}{R}$
      - B is wheel base
        - BG: 6m
        - MG: 4.88m
      - L is Lap of flange
        - $0.02\sqrt{hR+Dh} = L$

rawide

ing

We : $\frac{13\ (B+L)^2}{R}$

- h is depth of flange below top of rail in cm
- D is diameter of wheel in cm.

transition curve

$\frac{4}{5}$ of Calculated speed.

of setting-out horizontal curve in field

dolite method

ential offset method

d deflection method

chord offset method

17.6

Grade compensn

Broadgauge	0.04%/°C	or 70/R
Metre gauge	0.03½°C	or 52.5/R
Narrowgauge	0.02½°C	or 35/R

R is radius of curve in m.

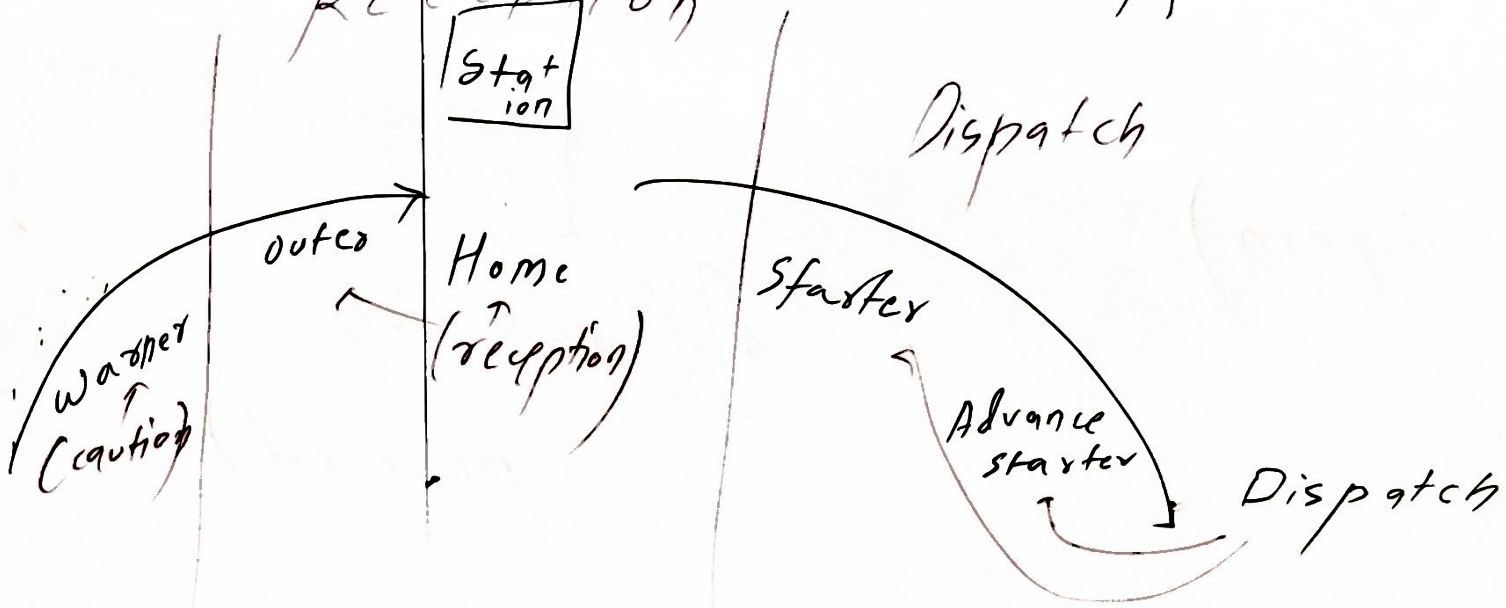
Signal

Signal Sequence

#2 times puche h.

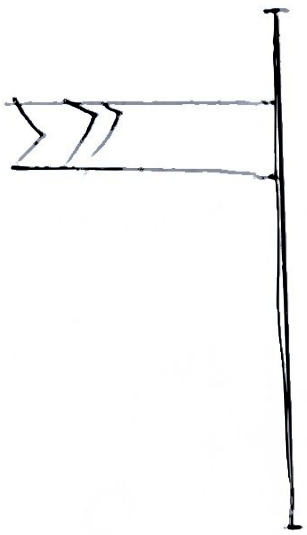
W - O - H - S - A

Reception

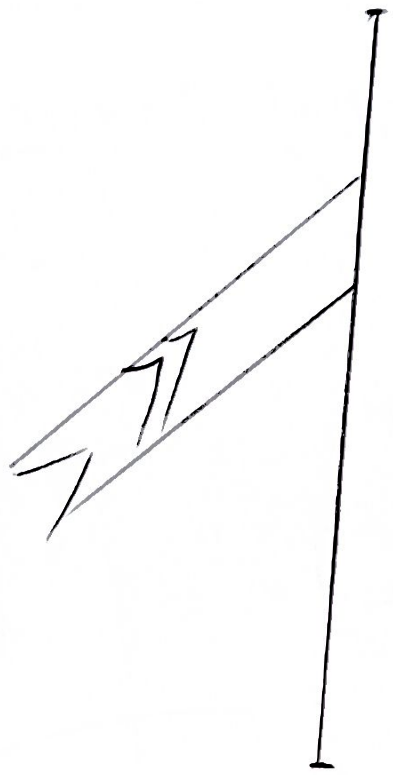


formation : gauge + formation → width in railway shoulder + drain.

10.2



Stop
(ON poin)



off position
(proceed)

Disk signal
!alternatename → shunting signal — used in station yard for shunting operation

a Mechanical devices used for interlocking the track or interlocking →

(1) point lock
(2) Treadler bar
(3) Detector
(4) Connecting devices.